

Learning Journey Assistant

Sprint Plan — Sprints 3, 4, 5 and 6

CSE5IDP Industry Development Project · Semester 2, 2026 · Group 3 · Jira project **IOLG**
Project owner: Dr Scott Mann · Repository: github.com/yetanotherpassword/LearningJourneyAssistant
Revision 5, 24 August 2026 — verified against the repository at commit `410abb6` ; reconciled against RFT 2026-CSE5IDP-S2-W (tender, submitted 23 August 2026)

Sprint 3 is **binding** and is what gets entered into Jira now. Sprints 4, 5 and 6 are **indicative** — recorded so dependencies are visible early, but not entered yet, because committing to detail eight weeks out is guesswork dressed as planning. Each is refined at the retrospective of the sprint before it.

Sprint numbering now follows the tender, not the internal plan

The tender commits publicly to **six sprints**, with Sprints 1 and 2 recorded as delivered actuals. The three remaining delivery sprints are therefore Sprints **3, 4 and 5**, and the 5–18 October fortnight is **Sprint 6** — which the tender already describes as "QA, security, documentation and handover... held partly as contingency."

This resolves the scrum-master problem raised in the previous revision. Six sprints across five members means every member presents at least once, exactly as §10 of the tender promises. No renegotiation with Ali is required — the commitment was already correct; only the internal document numbering was out of step.

1. Team

Names and component ownership from §10 of the tender. Sprint-plan slots M1–M5 are retired; work is now assigned to people.

Ref	Member	Component owned	Basis for sprint allocation
T1	Allan Campton	Software Architect	30+ years software engineering, defence and aerospace. Linux, Docker, Python, PostgreSQL. Currently holds the uncommitted lja/ codebase — which is why S3-1 can only be his. Takes the data model and gap engine.
T2	Ayesha Mosaddeque	Software Engineering	14+ years enterprise engineering; Salesforce, ex-AWS. CI/CD, automated testing, threat modelling, Terraform, Kubernetes. The only member with production CI and security experience — takes CI, scanning, and the Moodle extraction path.
T3	Istiaque Bhuiyan	Research and Development	Master of IT majoring in AI; research assistant at Murdoch. Takes the LLM layer, grounding, and generation features.
T4	Anup Tumbalam Gooty	Quality Assurance	Aviation IT service delivery; logistics management. Takes acceptance verification, the clean-machine rebuild, validation harnesses and the compliance checklist — assessable work that does not depend on prior software engineering depth.
T5	Sui Lung Tang	Risk Manager	Studying AI. Takes the dashboard views — the most accessible coding surface in the codebase, since they reuse an existing template and chart pattern — paired with Allan, plus the risk register.

Three inconsistencies in the submitted tender, for the implementation report

The tender is lodged and cannot be changed, but the implementation report should reconcile these rather than let a marker find them:

- **Team size.** §6, §7 and §3 say "six-person team" and "all six members" repeatedly, but §10 lists five (T1–T5). Five is correct. `devenv/env.sh` also says "all six of us" and should be corrected in the same commit that fixes the repo.
- **Effort arithmetic.** §6 costs 240 hours across six sprints — 40 hours per sprint. §7 states approximately 15 hours per member per week, which for five members over a fortnight is 150 hours per sprint. §6 and §8 agree with each other but neither agrees with the stated per-member commitment.
- **Role assignment.** The §3 risk register names T2 as extraction lead, product-owner proxy *and* QA lead, while §10 gives QA to T4. This plan follows §10.

2. Two tender commitments the code does not currently meet

These are not scheduling problems. They are gaps between what was formally offered on 23 August and what the codebase does, and both change the shape of Sprint 3.

Tender requirement	What was promised	What the code does
Req 4 — relative gap detection (Very High)	Gap detection "based on variability within an individual student's profile <i>rather than raw pass or fail thresholds</i> ", evaluated against hand-calculated cases with a sensitivity check.	<code>gap_detection.py</code> classifies against absolute thresholds of 50 and 65 — precisely the raw pass/fail mechanism the tender excludes. This is a direct contradiction, and it must be fixed rather than reconciled, because everything downstream consumes gap output. Now S3-6.
Req 2 — semantic mapping (Very High)	"Embeddings with LLM adjudication", with keyword matching "explicitly excluded as a production mechanism". Evaluated by a mapping report showing <i>similarity scores</i> and adjudication rationale per criterion.	There is no embedding stage anywhere in the repository. <code>silos_clustering.py</code> is LLM-only. It is genuinely semantic and it produces rationale, so the <i>spirit</i> of the requirement is met and keyword matching is not being used — but there are no similarity scores, so the promised evaluation evidence cannot be produced as written. Decide in Sprint 3, build in Sprint 4.

On requirement 2 there are two defensible responses and the team should pick one deliberately at Sprint 3 planning rather than drift. Either **add an embedding pre-pass** that proposes candidate SILO pairs by cosine similarity and hands them to the existing LLM for adjudication — which produces the similarity scores the evaluation criterion asks for, and would also constrain the coverage variance already observed across runs — or **document the deviation**, explaining that LLM-native clustering achieved the requirement's intent without an embedding stage, and adjust the evidence to rationale traces rather than scores. The first is more work and better evidence. The second is honest and cheap. Silently doing neither is the only bad option.

3. Calendar

Sprint	Dates	Capacity	Context
3	Mon 24 Aug – Sun 6 Sep	≈70%	Mid-semester break falls in week 2, from 31 Aug. Scope deliberately light. Take the days off — that is the lecturer's own instruction.

Sprint	Dates	Capacity	Context
4	Mon 7 Sep – Sun 20 Sep	100%	Sessions 8–9. Tender milestone: walking skeleton demonstrated end to end.
5	Mon 21 Sep – Sun 4 Oct	100%	Sessions 10–11. Feature freeze declared at the Sprint 5 review.
6	Mon 5 Oct – Sun 18 Oct	—	No engineering. Implementation report, sprint compendium, handover package, final presentation (12 Oct), submission Sunday 18 October. Still a real sprint: scrum master, ceremonies, burndown, presentation.

4. Sprint 3 — binding · Mon 24 Aug - Sun 6 Sep · 44 points

Sprint goal

Any team member can clone the repository and run the whole pipeline on their own machine; gap detection does what the tender says it does; and no gap report can be produced from unreviewed LLM clustering without saying so.

Behind that goal, as at 24 August: **the code has now landed.** Commit `eb93a01` brought `python/lja/` (22 modules — the LLM layer, the Excel loader, SILO clustering, gap detection, gap evidence, the CLI and the dashboard) and `python/tests/` (9 files) into `main`. Verified from a clean clone: the full suite runs green at **69 tests**, and a full-history secret scan is clean. The bus factor of one is closed and tender requirement 11 is now achievable. This was the single largest risk in the project and it is gone.

Three things follow from *how* it landed, and they reshape this sprint rather than reduce it.

- **It went in as one direct commit to `main`, not five reviewed pull requests.** The code is safe, but no peer review happened, and none can now happen through a PR that has already merged. The tender's §3 governance clause promises peer review before merge, and individual contribution is graded partly on review participation. S3-2 is therefore *reshaped*, not deleted — four members each take a module and review it retrospectively, raising findings as issues instead of PR comments. S3-3 prevents the situation recurring.
- **Nothing independently verified it.** A clean-machine rebuild by someone who did not write the code is the acceptance criterion, and it has not been done by a team member. S3-4 is unchanged and is now the most important remaining item in the sprint.
- **The READMEs are already drifting.** They describe 54 tests; there are 69. Small, but it is the same failure mode as before — documentation asserting something about the code that nobody checked. Folded into S3-4 rather than given its own ticket.

Still absent from `main`: `.github/workflows/` (no CI exists at all) and `docs/compliance-checklist.md`. Both were already in this sprint and both stand unchanged. `gap_detection.py` still carries `DEFAULT_LOW_THRESHOLD = 50.0` and `DEFAULT_HIGH_THRESHOLD = 65.0`, so the tender requirement 4 contradiction in §2 above is unaffected. There is still no embedding stage anywhere in the codebase.

Capacity note: S3-1 is complete on day one, which banks 5 of this sprint's 44 points immediately. The strengths dashboard view has been pulled forward from Sprint 4 to use the freed capacity.

Ref	Summary	Type	Assignee	Pts	Acceptance criteria
S3-1	Land the LJA Python package and test suite in <code>main</code>	Task	Allan	5	DONE — commit <code>eb93a01</code>, 24 Aug. 22 modules and 9 test files in <code>main</code> ; 69 tests pass from a clean clone; full-history secret scan clean; <code>output/</code> and <code>.env</code> confirmed ignored. Enter it in Jira and close it immediately — it is work done inside the sprint window and its points belong in the velocity record. The two acceptance criteria it did <i>not</i> meet (reviewed PRs, independent verification) are carried by S3-2 and S3-4.

Ref	Summary	Type	Assignee	Pts	Acceptance criteria
S3-2	Retrospective peer review of the landed codebase	Task	Ayesha x2, Istiaque, Anup, Sui Lung	5	Reshaped: the code went in as one direct commit, so there are no pull requests to review. Instead each member takes a module bundle, reads it, runs it, and raises findings as Jira issues linked to this task — not as Teams messages. Review still means running the code, not skimming it. Two purposes: it produces the review evidence the tender's governance clause promises, and it gets the other four members into a codebase currently understood by one person.
S3-3	CI pipeline and branch protection	Task	Ayesha	5	GitHub Actions on every PR to <code>main</code> : <code>ruff</code> , <code>pytest</code> , coverage report (tender promises 80% on core mapping and gap logic), secret scanning and dependency scanning (tender requirement 9, Mandatory). <code>main</code> protected: no direct pushes, one approving review, CI must pass. Badge in root README. Verified by confirming a deliberately failing test blocks a merge.
S3-4	Independent clean-machine rebuild verification	Task	Anup	2	Direct evidence for tender requirement 11, and now the most important remaining item in the sprint — the code is in <code>main</code> but no team member has independently rebuilt from it. Anup clones fresh on his own machine, creates the conda environment from the README alone, runs <code>python -m lja.cli</code> against the dataset and <code>pytest</code> . Both pass unassisted. Every point where the README was wrong or insufficient is raised as a bug — including the test count, which the READMEs give as 54 against an actual 69. All runtime dependencies are declared in <code>environment.yml</code> , so the conda path should work; the failure mode to watch for is anything he needs that the README does not tell him to install.
S3-5	Staff confirmation gate for LLM SILO clustering	Story	Istiaque	5	Three states per competency cluster: <code>pending</code> , <code>confirmed</code> , <code>rejected</code> . Persisted <i>alongside</i> <code>output/silo_clustering.json</code> — the cache is regenerable, the review decisions are not. <code>python -m lja.review</code> lists clusters with SILO members, sets state, attaches a note. <code>lja.cli</code> warns when a gap report is driven by <code>pending</code> clustering; <code>--allow-unconfirmed</code> required in scripts. A <code>rejected</code> cluster routes to a documented rework path, never a dead end. Unit test per transition.

Ref	Summary	Type	Assignee	Pts	Acceptance criteria
S3-6	Relative gap detection — replace absolute thresholds	Story	Allan	5	Closes the tender requirement 4 contradiction. Gap classification derives from variability within the individual student's own competency profile, not from fixed 50/65 cut-offs. Any remaining absolute value exists only as a configurable floor, from one source, never duplicated between <code>gap_detection.py</code> and the SQL. Existing tests re-expressed against the new semantics rather than pinned to old constants. The change and its rationale recorded as an ADR, because it alters what every downstream feature means.
S3-7	Current understanding level dashboard view	Story	Sui Lung (paired with Allan)	3	Per-student mastery across all competencies, not only gaps. Rendered from existing pipeline output — no new stage. Reuses the semantic palette in <code>dashboard/static/style.css</code> so chart and badge colours cannot disagree. Covered by the existing in-memory dashboard fixtures.
S3-8	Close the competency-mechanism architecture decision record	Task	Ayesha	3	Time-boxed to one day, hard stop. Load <code>competency_framework_cse5idp.csv</code> into the devenv Moodle, run SQL queries 3 and 4, record which returns rows in <code>sql/README.md</code> with the output that justifies it. If neither returns rows, record that too — it makes the bridging table the product rather than a workaround, which is a defensible architectural claim for the design report.
S3-9	Verify the Anthropic client against a live API key	Task	Istiaque	2	Evidence for tender requirement 10 (provider abstraction). The <code>AnthropicClient</code> path has never been run — only local Ollama has. One live structured-output call succeeds end to end; result recorded in <code>python/README.md</code> . No key committed.
S3-10	Close out the last stale branch	Task	Anup	1	Rescoped: Istiaque/Tender-Document was merged via PR #3 on 23 Aug and the tender is now in <code>main</code> . Only <code>docs/macros-setup-readme</code> (Ayesha's Apple Silicon setup guide, 16 Aug) remains unmerged — confirmed not reachable from <code>main</code> . Merge it or close it with a written reason. It is setup documentation, so it belongs in the handover.
S3-13	Strengths dashboard view	Story	Sui Lung	3	Pulled forward from Sprint 4 using the capacity freed by S3-1 landing early. High-attainment competencies as a first-class view, threshold-driven from S3-6's configuration rather than a second hardcoded number. Second of the four views tender requirement 5 promises.
S3-11	Draft the compliance checklist	Task	Anup	3	<code>docs/compliance-checklist.md</code> created from the proposal's admin checklist — legal, security, financial, technical, HR, communication, QA, documentation. Draft state acceptable this sprint; every row has an owner and a status. Completed in Sprint 5.

Ref	Summary	Type	Assignee	Pts	Acceptance criteria
S3-12	Move the risk register into Jira	Task	Sui Lung	2	The eight risks from tender §3 exist as a table in a Word document, which means they will not be reviewed. Recreate them as Jira issues or a linked page with likelihood, impact, control, owner and residual level, so they can actually be re-scored at each retrospective as the tender promises. Reassign the owners to match §10 rather than the tender's inconsistent risk table.

Sub-tasks

S3-1 landed as a single commit, so its five planned PR sub-tasks no longer apply. Only S3-2 decomposes. Sub-tasks carry no story points — the parent already carries the estimate, and pointing both double-counts the burndown — but they do carry individual names, which is where evidence of individual contribution comes from.

Parent	Sub-task — module bundle to review	Assignee
S3-2	lja/config.py and lja/llm/ — the provider abstraction	Ayesha
S3-2	lja/data/ — Excel loader and synthetic generator	Anup
S3-2	lja/model/ — clustering, gap detection, gap evidence	Ayesha
S3-2	lja/dashboard/ — app, templates, static	Sui Lung
S3-2	lja/cli.py and tests/	Istiaque

Workload distribution

Member	Items	Points
Allan	S3-1 land code (5, <i>done</i>) · S3-6 relative gap detection (5)	10
Ayesha	S3-3 CI and scanning (5) · S3-8 competency ADR (3) · two module reviews (2)	10
Istiaque	S3-5 confirmation gate (5) · S3-9 Anthropic verification (2) · one module review (1)	8
Anup	S3-4 rebuild verification (2) · S3-10 stale branch (1) · S3-11 compliance draft (3) · one module review (1)	7
Sui Lung	S3-7 understanding view (3) · S3-13 strengths view (3) · S3-12 risk register (2) · one module review (1)	9
Sprint 3 total (5 already banked on day one)		44

On the 10 / 10 / 9 / 8 / 7 spread

This is deliberately not flat. The team has two members with 14 and 30 years of professional software engineering, one research-oriented member, and two whose stated backgrounds are outside software development. Assigning identical point loads across that spread would mean either under-using the two senior engineers or handing sprint-critical work to members who would need the whole fortnight to attempt it — and the sprint would fail on the second.

What is distributed evenly is **assessable contribution**. Every member owns merged work, carries at least one review, and has something concrete to present. Anup's S3-4 is the acceptance gate for the single most important story in the sprint. Sui Lung's dashboard view is real user-facing code paired with the architect, not documentation busywork. Both are defensible in a scrum-master presentation, which matters because individual contribution is graded.

Decisions to put to Scott inside Sprint 3

Both block Sprint 4 stories. Raise them in the Teams channel this week — the project owner has a day job and may take a week to reply.

1. **Trajectory semantics.** The supplied workbook has no dates and no attempt timestamps anywhere — 1650 rows, 11 assessments, no time dimension. "Progress trends" can only mean progression across year levels (CSE10OF → CSE2ALG → CSE3CAP), not a time series. Confirm before S4-5 is built on it.
2. **Export field set.** Confirm what the department's longitudinal and A/B evaluation needs per record before S5-1 defines the schema. The tender lists this as Mandatory, and the point of the export is that runs from different times are comparable — changing the schema after data exists is expensive.

Sprint 3 review demo

Anup — who did not write the code — drives the demo from his own clean clone: CLI run, the new understanding-level view, a cluster moved from `pending` to `confirmed` with the badge changing, and gap classifications produced by the new relative logic rather than the old thresholds. **A demo on Allan's laptop does not meet the sprint goal**, however well it runs.

If the break costs more time than planned

Drop S3-11 first, then S3-13, then S3-12. **S3-2 through S3-6 are non-negotiable** — S3-1 is already done, and without the rest the sprint goal is not met regardless of what else ships.

5. Sprint 4 — indicative · Mon 7 Sep - Sun 20 Sep · 50 points

Not entered into Jira yet. Refine at the Sprint 3 retrospective. **Goal:** the walking skeleton runs end to end on both data paths, and the first student-facing generated artefact exists and is provably grounded. This is the tender's "walking skeleton demonstrated end to end" milestone.

Ref	Summary	Assignee	Pts	Note
S4-1	Moodle vertical slice, time-boxed to this sprint	Ayesha	8	One subject, seeded rubric fillings, SQL query 2 through a Python reader into <code>lja_criterion_score</code> , consumed by the same gap engine as the Excel path. If the Moodle path needs its own gap engine the abstraction is wrong, and that is worth finding out now. Hard stop at the sprint boundary.
S4-2	Table prefix as configuration, not a constant	Ayesha	2	Our own two environments already disagree — <code>mdl_</code> versus <code>devenv's m_</code> .
S4-3	Reusable grounding validator	Allan	5	Generalise <code>_validate_coverage()</code> — which already caught a live model silently dropping three of thirteen SILOs — into <code>lja/llm/grounding.py</code> , usable by any generated artefact. Prerequisite for S4-6.
S4-4	Trajectory model	Allan	5	Attainment ordered along a <i>declared</i> subject-sequence configuration, not parsed from the course code. The no-timestamps constraint stated in the docstring and on the page.
S4-5	Progress-trends dashboard view	Sui Lung	5	Completes all four views tender requirement 5 promises. Strengths moved to Sprint 3 (S3-13).
S4-6	Learning-plan generation with grounding tests	Istiaque	8	Grounding tests are part of the story, not a follow-up. Every SILO, subject, assessment and competency name in the output must appear in the input, or the build fails — tender requirement 6.
S4-7	Embedding pre-pass, or documented deviation	Istiaque	2	Whichever option §2 of this document resolves to. If embeddings are added, the deliverable is candidate pairs with cosine scores handed to the existing LLM adjudication. If not, a written deviation note against tender requirement 2.
S4-8	Deterministic gap validation harness	Anup	5	Direct evidence for tender requirement 4: hand-calculated expected results across synthetic profiles, asserted against the new relative logic. This is the "validation worksheet" the tender promises.
S4-9	Security pass 1	Anup	3	Confirm <code>lja_reader</code> is genuinely read-only, token capabilities match documentation, no unresolved high-severity findings from the CI scanners.
S4-10	Dev-mode student selector	Sui Lung	2	Full authentication stays out of scope and is recorded as a known limitation.
Total — Allan 10, Ayesha 10, Istiaque 10, Anup 8, Sui Lung 7			45	

6. Sprint 5 — indicative · Mon 21 Sep - Sun 4 Oct · 50 points

Goal: the last feature ships, then everything becomes evidence and hardening, so the 5–18 October fortnight is free for the report and the final presentation.

Ref	Summary	Assignee	Pts	Note
S5-1	Export layer for longitudinal and A/B evaluation	Allan	8	Tender requirement 7, Mandatory . Per-student and cohort-level attainment, gap classification, trajectory and plan, as CSV and JSON. Versioned schema, because the comparison depends on exports from different runs being diffable. -- anonymise using salted stable pseudonyms.
S5-2	Architecture and handover document	Allan	2	Including the rubric-fills-via-database caveat stated plainly — it is not API-mediated and the handover has to say so.
S5-3	Feedback attribution layer	Ayesha	5	Every finding points at a source row and a character span; a paraphrase is never presented as fact. Degrades cleanly when the comment is one of the 45 templated strings, which most are. Closes the proposal objective the pipeline currently skips entirely.
S5-4	Coverage to 80% on mapping and gap logic	Ayesha	3	The tender commits to this number in its evaluation matrix. Measure, then close the gap.
S5-5	Moodle path README truth pass	Ayesha	2	Forward-looking language rewritten as what is actually true.
S5-6	Study-strategy recommendations	Istiaque	8	Tied to gap <i>classification</i> , so an isolated gap and a persistent cross-subject gap get structurally different advice. Every recommendation cites the competency and its evidence. Last feature built.
S5-7	Extend the grounding suite to cover S5-6	Istiaque	2	
S5-8	Full regression pass	Anup	3	Across every must-have.
S5-9	Complete the compliance checklist	Anup	5	Every item closed or explicitly deferred with a written reason.
S5-10	Demo rehearsal from a clean clone	Anup	2	Against the running system, not slides.
S5-11	Surface plans and strategies in the dashboard	Sui Lung	5	
S5-12	Feedback evidence panel under each gap	Sui Lung	3	Renders S5-3's attributed feedback with its source visible — tender requirement 5's traceability clause.
S5-13	Threshold sensitivity chart and risk register close-out	Sui Lung	2	Student counts per band across a sweep. Quantified evidence for the report.
Total — 10 points per member			50	

7. Sprint 6 — indicative · Mon 5 Oct - Sun 18 Oct

No engineering. The tender describes this sprint as "QA, security, documentation and handover... held partly as contingency", and that is exactly what it should be. Backlog: implementation report (2000 words per student), sprint compendium (six reflections), handover package acceptance, final demonstration on 12

October, submission Sunday 18 October. It still carries a scrum master, ceremonies, a burndown and a graded presentation.

Any feature carried into Sprint 6 is paid for out of report-writing time. That is why the freeze is declared out loud at the Sprint 5 review.

8. What was cut, and why it is on the record

Cut	Rationale	How it is reported
Adaptive quiz generation	Pre-agreed descope priority 1, and the tender itself names it "first descope candidate" at Medium importance — the only requirement below High. Highest complexity-to-value ratio of the AI features, and the one most likely to look plausible while quietly not being grounded.	Named in the implementation report as a planned descope with reasoning. Not silently omitted.
Generalised Moodle production loader	Reduced to one time-boxed vertical slice (S4-1).	Documented design plus spike evidence, and the rubric-fills caveat stated plainly in the handover.
Confirmation UI on the dashboard	CLI review only. The gate itself is not cut — only its web interface.	Known limitation.
Authentication	Dev-mode student selector only.	Known limitation.

Not cut under any schedule pressure: the extraction → clustering → gap-detection spine, the four dashboard views, the confirmation gate, the export, and learning plans. Those are the tender's Very High and Mandatory requirements. A working dashboard with honest gap detection is a defensible MVP on its own even if every generative feature above it is cut.

9. Definition of Ready and Definition of Done

Ready — before an item enters a sprint

- Acceptance criteria written in testable terms, not aspirations.
- Story points estimated by the team, not by the author alone.
- An owner identified, and the owner agrees.
- Dependencies on other items or on a project-owner decision known and recorded.
- Attached to an epic.

Done — before an item is closed

- Merged into `main` via a pull request reviewed by someone other than the author.
- `pytest` passes; new logic carries unit test coverage.
- The relevant bundle README updated if setup or behaviour changed — not deferred to "docs later".
- No secrets committed; `.env.example` updated if new configuration was added.
- **Demoable** — reachable from the dashboard or a documented CLI entry point.
- Acceptance evidence attached to the Jira issue, as the tender's §3 governance clause promises.
- The Jira issue moved to Done *as the work completes*, not the night before the presentation.

10. Process items — currently the weakest part of the project

- **Commit spread.** Three of five members have committed; Allan holds 15 of 19 commits. Individual contribution is assessed and `git log` is the artefact. S3-1's five-PR split and S3-2's review allocation exist partly to create that evidence.
- **One source of truth.** Jira (IOLG) is the graded surface and the tender names it "the single source of work and evidence". Reference Jira keys in commit messages (`Refs IOLG-nn`) so the burndown, the ticket and the code change form one chain.
- **Burndown honesty.** Move tickets as work completes. Sprint 3's ticket movement is the first burndown anyone outside the team will see. An honestly explained bad chart costs nothing; a fabricated good one is not acceptable.
- **Sprint review with the project owner at every boundary,** demoed from the running system rather than slides.